

FREE180nm MODEL LIBRARY USER GUIDE

Version: 1.1 **Technology:** 180nm Bulk CMOS & GFM_MCU IO Library **Purpose:** Support Analog and Mixed-Signal circuit design and simulation in LTspice for educational and research purposes.

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Project Information

Item	Information
Library Name	FREE180nm Model Library
Version	1.1
Technology	180nm Bulk CMOS & GF180MCU IO Library
Spice Tool	LTSpice (Version: LTspice 24.0.12)
Author	Max
Email	mr.max1186@gmail.com
Date	2026-09-19
Purpose	Analog and Mixed-Signal circuit design and simulation using LTspice
License / Distribution	Provided free of charge for educational, research, and experimental purposes

Contact: For questions, feedback, bug reports, or technical discussions related to the FREE180nm Model Library, please contact the author at mr.max1186@gmail.com

1. GENERAL INTRODUCTION & MODEL FILE STRUCTURE

To use this library in an LTspice schematic, you need to declare the main model file at the beginning of the netlist or directly include the required files using the `.include` command.

Standard File Header

Use the following header at the beginning of the main model file:

```
=====
=
File Name:   FREE180nm.model_TT
Purpose:     Create to Study Analogic Circuit Design on LTspice
Target Device: PTM 180nm Bulk / GFM_MCU IO Buffer
Author:      max1186
Date:        2026-09-19
Version:     1.1
Notes:       Integrated with encrypted PTM 180nm and GFM_MCU libraries.
              Developed for accurate transient, DC analysis, AC simulation
=====
=

.include 'PTM180nm_bulk_TT.txt'
.include 'models_gfmcu.lib'
```

2. BASIC COMPONENT LIST (CORE DEVICES)

The library provides transistor models classified according to their voltage rating, allowing flexible design ranging from low-voltage digital/analog core circuits to high-voltage interface blocks.

Device Model			
Name	Type	Voltage Rating	Typical Applications
ptm180_nmos1p8	NMOS	1.8 V	Core logic circuits, transmission gates, basic inverter output stages.
ptm180_pmos1p8	PMOS	1.8 V	Core logic circuits, pull-up networks.
gfmcu_pmos3p3	PMOS	3.3 V	Cross-coupled loads for level shifters, high-voltage auxiliary loads.
gfmcu_nmos3p3	NMOS	3.3 V	Protective cascode transistors, 3.3 V IO output driver circuits.

3. PASSIVE COMPONENTS & SPECIAL BJT DEVICES

These components are intended for advanced circuit designs such as bias-generation circuits and Bandgap Reference (BGR) circuits.

3.1. Vertical PNP BJT

Model: gfmcu_vnpn_5x5

Application: Used as a core cell in BGR design to generate the base-emitter voltage difference ΔV_{BE} through an emitter-area ratio, for example a ratio of 1: 8.

Syntax:

X<Name> <Collector> <Base> <Emitter> gfmcu_vnpn_5x5 mp=<val>

3.2. Poly-Silicon Resistors

P-Poly Resistor

Model: gfmcu_ppolyf_u

P-Poly resistor, mainly used for feedback networks and voltage-divider circuits.

N-Poly / N-plus Resistors

Models: gfmcu_npolyf_u / gfmcu_nplus_u

N-Poly / N-plus resistors, used to generate PTAT current:

$$I_{PTAT} = \frac{\Delta V_{BE}}{R}$$

4. DEVICE DECLARATION RULES & GEOMETRICAL PARAMETERS

When calling customized subcircuits or MOSFET devices, the general SPICE syntax is as follows.

4.1. General Syntax for MOSFETs & Resistors

X<Name> <Node1> <Node2> <Node3> <Node4> <Model_Name> wp=<val> lp=<val>
np=<val> mp=<val>

4.2. Parameter Definitions

Parameter

Description

wp	Channel width / segment width. Unit: <i>m</i> .
lp	Channel length / segment length. Unit: μ .
np	Number of transistor fingers.
mp	Multiplier for the number of devices connected in parallel.
ms	Number of segments for resistors.

5. STANDARD BUILDING BLOCKS

5.1. 1.8 V Inverter (inv1)

```
.subckt inv1 A VDD VSS YB
```

```
XM3 YB A VDD VDD ptm180_pmos1p8 wp=1u lp=0.18u np=1 mp=1  
XM0 YB A VSS VSS ptm180_nmos1p8 wp=0.5u lp=0.18u np=1 mp=1
```

```
.ends inv1
```

5.2. 3.3 V Inverter (inv1_33)

```
.subckt inv1_33 A VDD VSS YB
```

```
XU1 YB A VDD VDD gfmcu_pmos3p3 wp=1u lp=0.3u np=1 mp=1  
XU2 YB A VSS VSS gfmcu_nmos3p3 wp=0.5u lp=0.3u np=1 mp=1
```

```
.ends inv1_33
```

6. BASIC MODEL CHARACTERIZATION TESTS

Before using the MOSFET models in larger circuit designs, the integrity and electrical behavior of the models can be evaluated using the following three characterization tests.

6.1. $I_D - V_{DS}$ Output Characteristics

Purpose

Measure the drain current (I_D) as a function of drain-source voltage (V_{DS}) at different gate-source voltage (V_{GS}) levels.

This test is used to examine the linear and saturation regions of the MOSFET.

LTspice DC Sweep Command

Example:

```
.dc vdd33 0 3 0.05
```

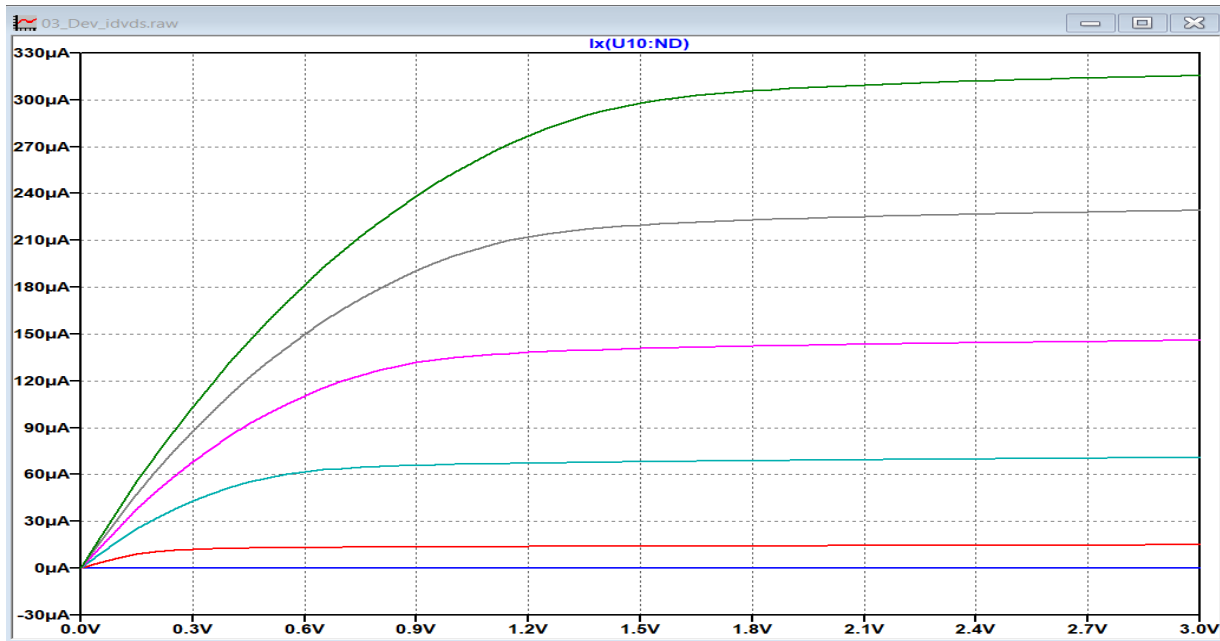
The sweep performs the following:

- Sweeps V_{DS} from 0 V to 3 V.
- Uses a V_{DS} step size of 0.05 V.
- Sweeps V_{GS} from 0 V to 3 V.
- Uses a V_{GS} step size of 0.5 V.

```
*
E:\00_Semi_Course\Model\Free180nm_PDK_20251023\00_Transfer_to_student\LAB
\Lab0\03_Dev_idvds.asc
vdd33 VDD33 0 3.3
.param pvgn33=0
.step param pvgn33 0.5 3 0.5
.dc vdd33 0 3 0.05
.inc E:\00_Semi_Course\Upload\Model\FREE180nm.model_TT
.param pvdd33=0
vvg33 vgn33 0 {pvgn33}

XU10 VDD33 VGN33 0 0 gfmcu_nmos3p3 wp=1u lp=0.5u np=1 mp=1

.lib FREE180nm.model
.backanno
.end
```



6.2. $I_D - V_{GS}$ Transfer Characteristics

Purpose

Measure the variation of drain current (I_D) as a function of the control voltage V_{GS} while keeping V_{DS} constant.

For example, V_{DS} can be fixed at:

- $V_{DS} = 3.3 \text{ V}$

This test can be used to evaluate:

- Transconductance (g_m).
- Off-state leakage current (I_{off}).

LTspice DC Sweep Command

Example:

```
.dc Vgs 0 3 0.01
```

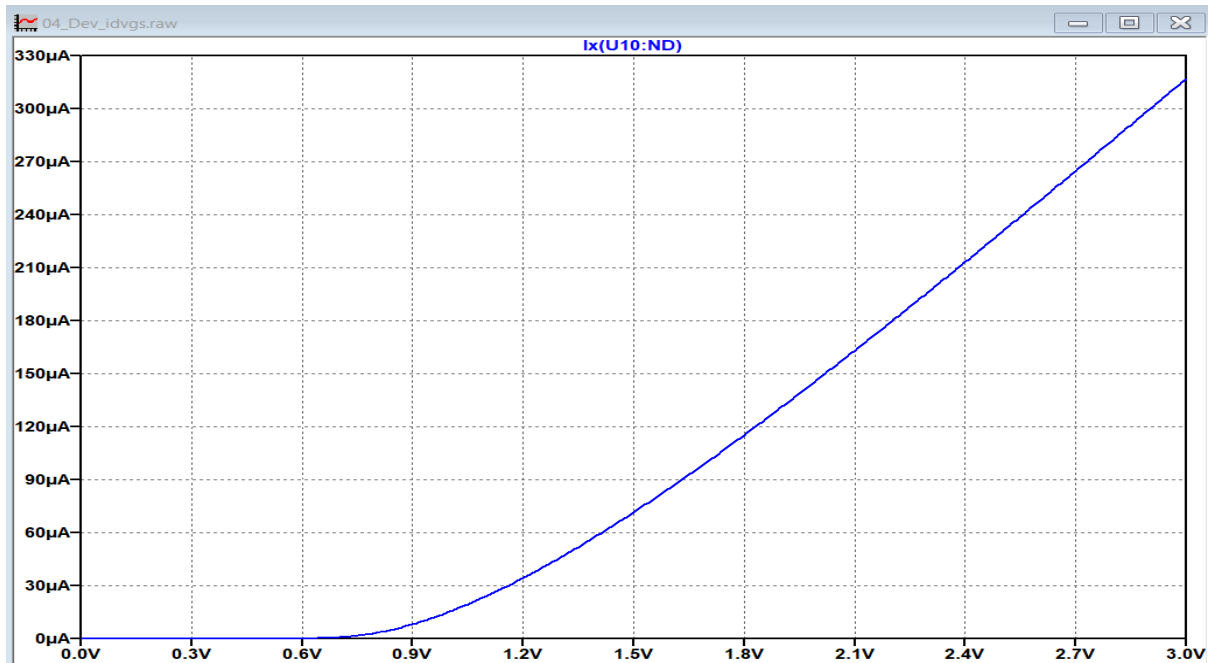
```
* LTspice: Simulation ID-VGS
* Diễn Đàn Học Vi Mạch Cơ Bản

E:\00_Semi_Course\Model\Free180nm_PDK_20251023\00_Transfer_to_student\LAB\Lab0\04_Dev_idvgs.asc
.inc E:\00_Semi_Course\Upload_git_Face\Model\FREE180nm.model_TT

vvd33 VDD33 0 3.3
.param pvgn33=0
.dc vvgn33 0 3 0.01
.param pvdd33=0
vvgn33 vgn33 0 {pvgn33}

XU10 VDD33 VGN33 0 0 gfmcu_nmos3p3 wp=1u lp=0.5u np=1 mp=1

.lib FREE180nm.model
.backanno
.end
```



6.3. Threshold Voltage (V_{th}) Measurement

Purpose

Determine the MOSFET threshold voltage accurately using a specified reference-current method.

For example, the threshold voltage can be measured at:

$$I_D = \frac{W}{L} \times 100 \text{ nA}$$

Alternatively, the .meas command can be used in LTspice.

Example Measurement Command

```
.meas dc vth FIND v(g) WHEN i(Vd)=1e-7
```

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Source and Attribution

The library integrates or is based on freely available models and resources, including:

- **GF180MCU PDK (Free / Open PDK resources)**
- **Predictive Technology Model (PTM) 180nm models**

For the GF180MCU PDK electrical specifications and related SPICE model information, refer to the official documentation:

[GF180MCU PDK – Analog SPICE Electrical Specifications](#)

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REVISION HISTORY

Version	Release Date	Author	Description of Changes / Key Updates
1.0	2026-04-30	Max	• Initial release of the FREE180nm Model Library. • Integration of PTM 180nm bulk CMOS and GFM_MCU IO buffer subcircuits. • Setup of standard layout parameter definitions (wp, lp, np, mp).
1.1	2026-09-19	Max	• Major Update: Converted all geometrical parameters and internal wrapper units across the entire model package (models_gfmcu.lib, PTM*.txt model) to Standard SI Meter (m) units. • Standardized instance parameter mapping in LTspice (e.g., explicit use of scientific notation or unit suffixes such as 1e-6 or 1u for micrometers).